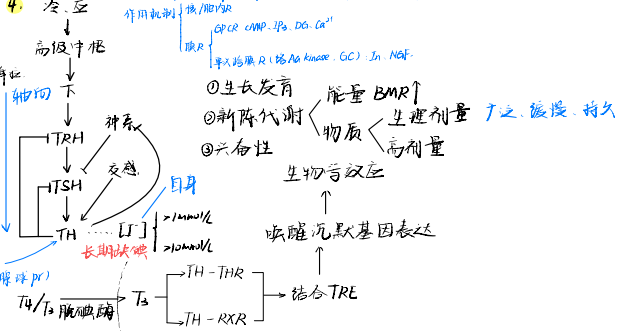
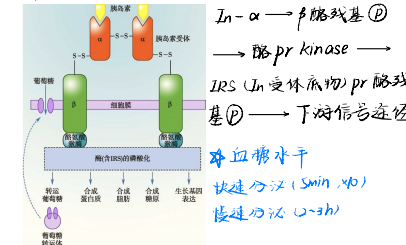
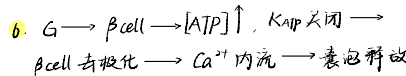
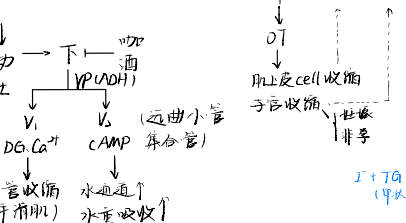
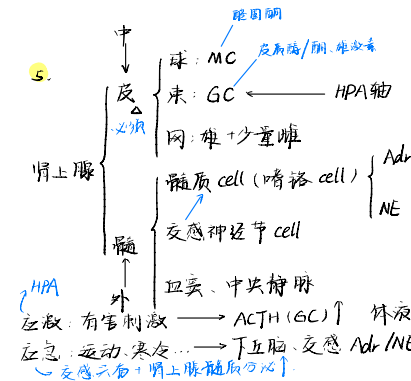
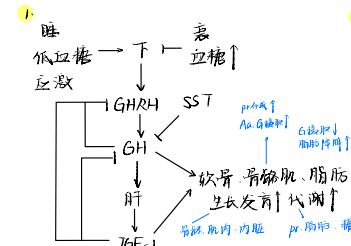
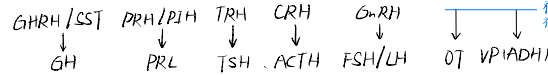


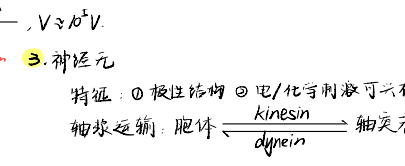
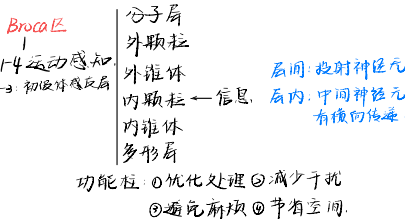
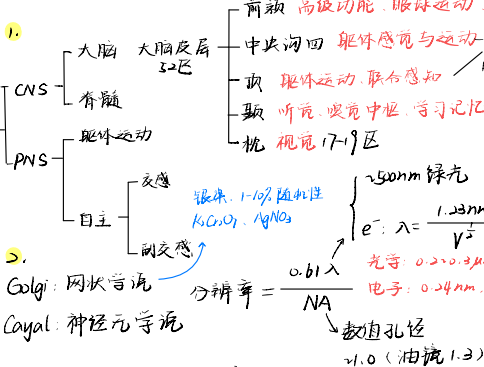
内分泌 作用: 维持、调节、促进、调节 形式: 运种、内、自、旁 种类: 胺(儿、胃、甲、醛)、多肽/Pr、In、VP/ADH、OT、类固醇(孕、雄、雌、反、甲状腺)



	甲亢 (甲状腺功能亢进症)	甲减 (甲状腺功能减退症)
产热	BMRI患者体温↑, 怕热, 多汗	相反
蛋白质代谢	蛋白分解↑→消瘦	蛋白合成↓, 组织间黏蛋白↑→黏液性水肿
糖代谢	进食后血糖↑↑ (糖尿), 但血糖正常或偏低	血糖正常或偏低
脂肪代谢	胆固醇分解↑→血胆固醇↓	胆固醇、甘油三酯↑
神经系统	CNS兴奋性↑→易怒烦躁、失眠、注意力不集中、肌颤等	相反
生长发育	正常	幼儿智力低下, 身材矮小 (呆小症)
循环系统	心脏正性变→心悸、心肌肥大甚至心衰; 血管舒张→脉压↑	心率减慢, 心音无力, 中老年血压升高

Feature	Nervous System	Endocrine System
Signals	electrical impulses (action potentials)	chemical impulses (hormones)
Pathways	transmission by neurons	transported by blood
Speed of information	fast	slow
Duration of effect	short lived	short or long lived
Type of action and response	voluntary or involuntary	always involuntary
Target	localized (cells connected to neuron)	often distant (many cells can be effected)

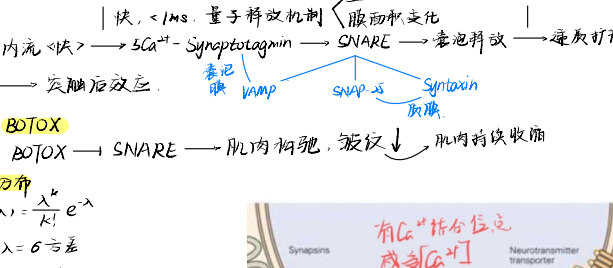
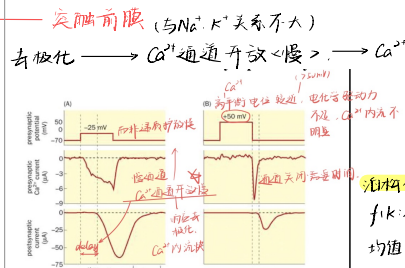
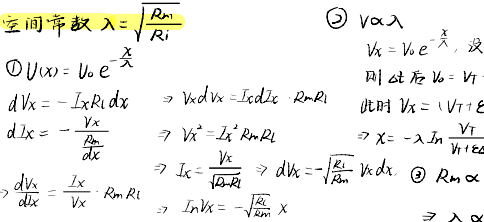
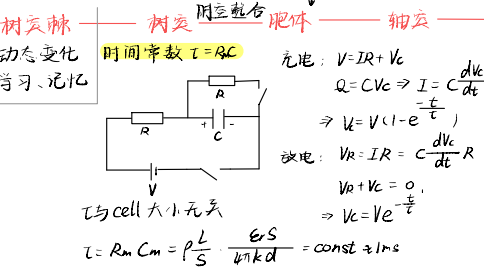
神经系统的结构



神经胶质细胞

功能: 支持、回收递质、保持离子环境、分隔、营养、清除
类型: 少突、Schwann、星形 (CNS) 星形、小胶质 (PNS)

神经信号的流动



感觉简介

感觉的编码

爆发式AP: 频率/间隔, 不应期限制下的频率上限。喂cell 连续变化的AP 视觉、听觉、味觉

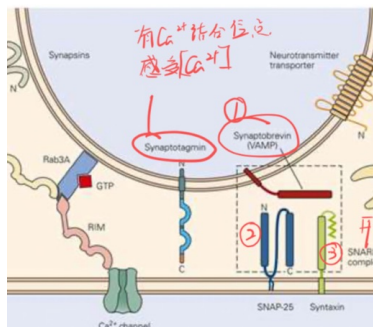
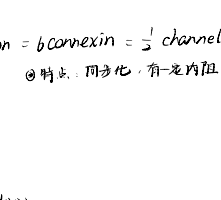
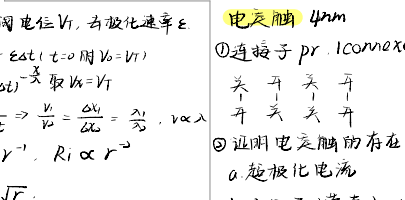
感觉的感受特点

高度敏感性: 视<单光子>, 听<纤毛热运动>, 嗅<单个信息素>
极大可感知范围
感觉用信号传导: ① 捕捉单一刺激 ② 高效放大 ③ 为神经传递提供信号

Gpr 视觉、嗅觉、味觉(甜、苦)、高放大低反应
直接听觉、触觉、味觉(咸、酸)、低放大高反应

感觉通路的拓扑映射

视网膜、躯体感知、声频率



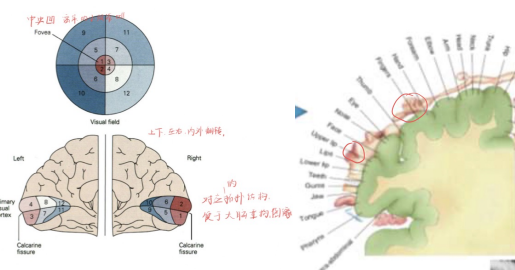
感觉的辨别

颜色: 色彩抵抗学说: B-Y, R-B cell (视锥, 相反的反应)

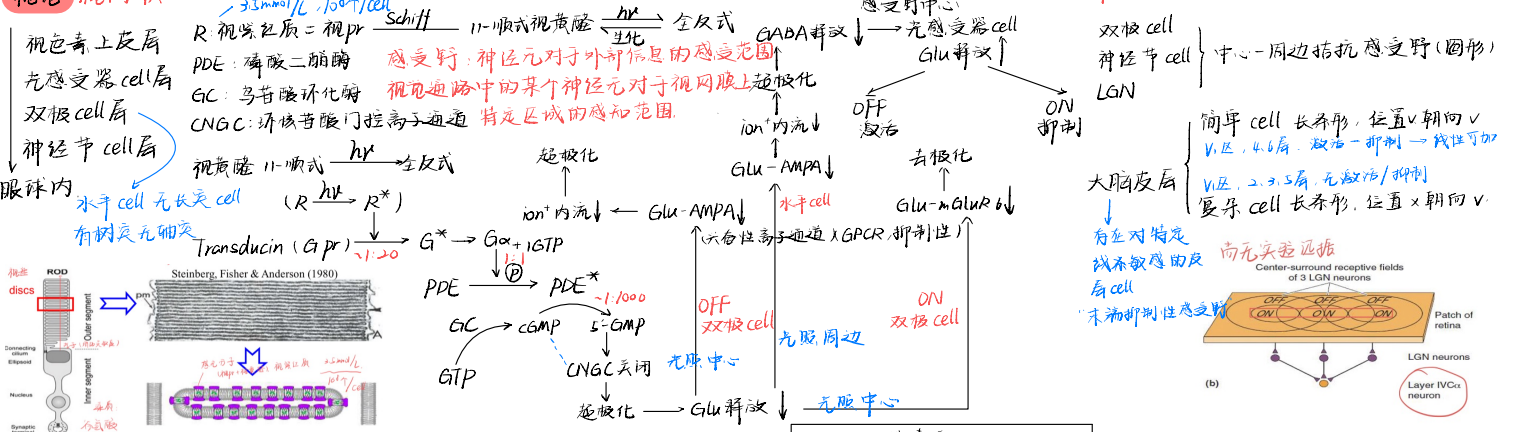
嗅觉: 嗅小球连接同类型受体的嗅cell

同一种气味分子——多种受体
多种气味分子——一种受体——一种cell 组合编码

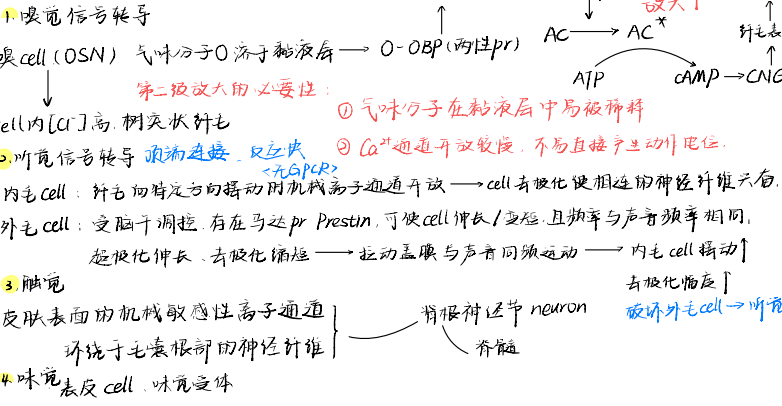
应急反应: 交感——肾上腺髓质活动增加, ADR/NE浓度增加使肺通气增加, 心脏收缩力增加, 心率加快, 血压升高, 糖原脂肪分解提供能量, 使血液流向骨骼肌和肝脏。应激反应: 下丘脑——垂体——肾上腺皮质轴被激活, 糖皮质激素分泌量增加, 促进糖原的合成, 保证向心脏和脑提供能量, 同时可以长时间保持较高血压。应急反应和应激反应通常不能拆分, 应急反应是紧急短时刺激时机体的对抗机制, 而应激反应则是长时间机体对刺激的适应和对抗过程, 两个机制经常一起作用



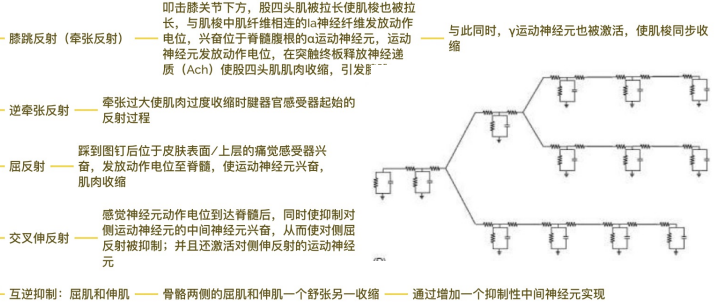
视觉 视网膜 视神经 外侧膝状体 LGN 大脑皮层



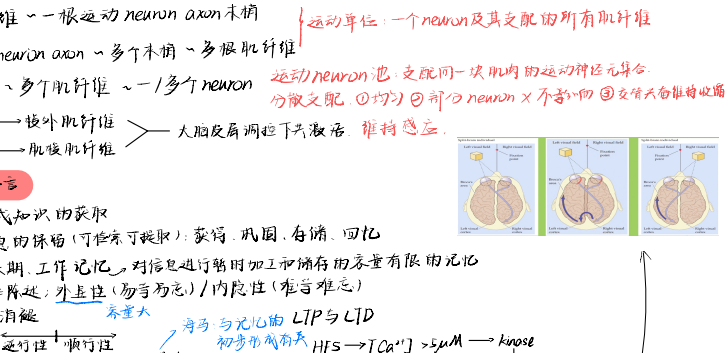
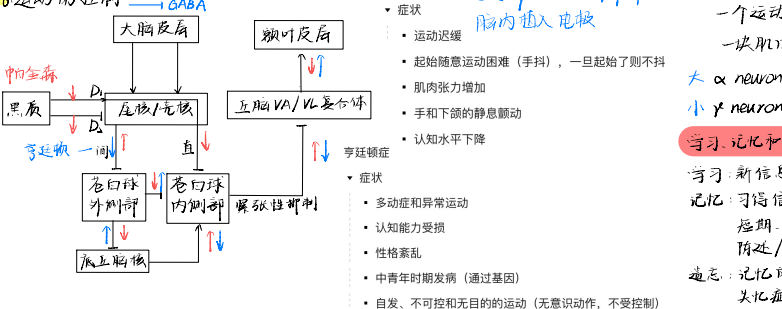
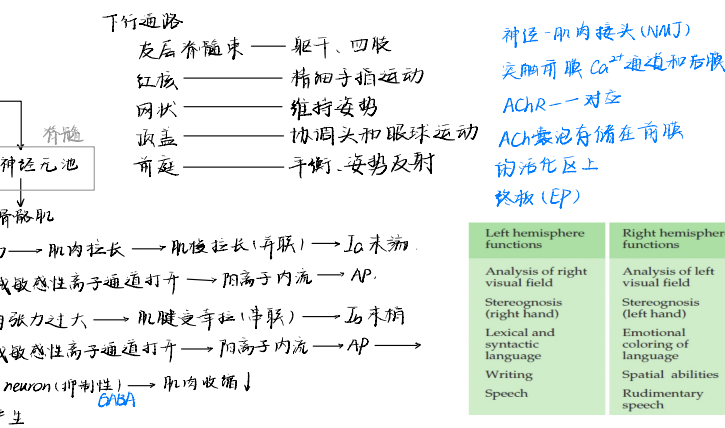
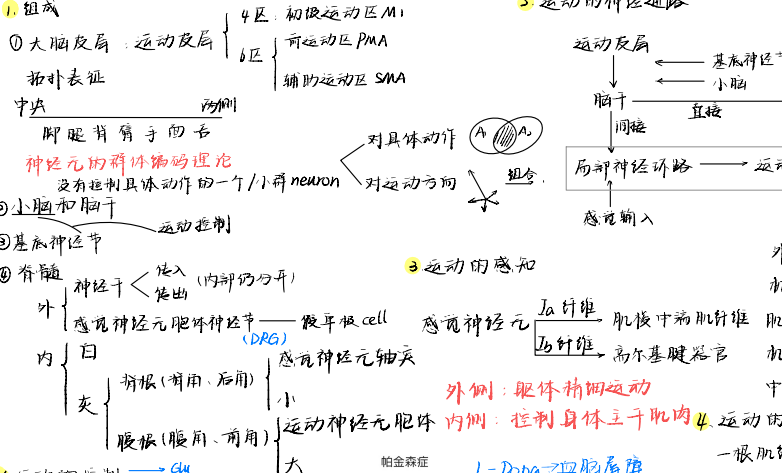
嗅觉、听觉、味觉、触觉



3. 常见的反射



躯体运动系统



自主神经系统

